

HU Collection

TMUA Mock Set J Paper 2

Worked Solutions

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Answer key

Q	1	2	3	4	5	6	7	8	9	10
Ans	C	D	E	F	A	B	E	F	A	D

Q	11	12	13	14	15	16	17	18	19	20
Ans	D	C	F	G	B	E	C	B	A	G

Each solution gives the intended route and names the trap the question was built around. Every question is pitched at the hard end of the TMUA, and they are loosely ordered by increasing difficulty.

Question 1. Answer C.

I denies the conclusion of an implication and so denies its hypothesis: valid. II runs the implication backwards and is invalid, since $n = 3$ makes both premises true and the conclusion false. III: the second premise rules out “ n^2 is odd”, so the first premise leaves only “ n is even”: valid. IV is valid directly. V has the same flaw as II: the two implications chain to “if n is a prime greater than 2 then $n^2 + 1$ is even”, and knowing the conclusion says nothing about the hypothesis; $n = 9$ makes all three premises true. So I, III and IV. Validity is about whether the conclusion follows, not about whether the premises are true: in III the first premise happens to be true for every integer, but that is not what makes the argument valid.

Question 2. Answer D.

Translate each fact. (1) says Mechanics implies Pure. (2) says: if a student does not take Decision, then they do not take both Statistics and Mechanics; equivalently, Statistics and Mechanics together force Decision. (3) says Decision implies Statistics. I follows from (1). II: Aisha takes Mechanics and not Decision, so if she took Statistics, (2) would force Decision; hence she does not. III: Mechanics gives Pure by (1), and Decision gives Statistics by (3). IV does not follow: a student taking only Decision and Statistics is consistent with every fact. V does not follow either: a student taking all four modules satisfies (1), (2) and (3), and (4) is only about Aisha. So I, II and III.

Question 3. Answer E.

I: if a , ar , ar^2 are in arithmetic progression then $2ar = a + ar^2$, so $a(r - 1)^2 = 0$: either every term is 0 or $r = 1$, and the sequence is constant either way. II: $\log(ar^{n-1}) = \log a + (n - 1)\log r$. III: $2^{a+(n-1)d} = 2^a (2^d)^{n-1}$. IV fails: 1, 2, 4 plus 1, 3, 9 gives 2, 5, 13. V: with common difference d , $u_{n+1}^2 - u_n^2 = d(2u_n + d)$, and this increases by $2d^2$ from each n to the next, so it is constant only if $d = 0$. VI fails: $u_1 = S_1 = 3$, while $u_n = S_n - S_{n-1} = 2n$ for $n \geq 2$, giving 3, 4, 6, 8, ...; the formula $2n$ breaks at $n = 1$ because S_0 is 0, not 1. Four statements are true.

Question 4. Answer F.

Lines (1) to (5) are all correct: the substitution maps $[0, 2]$ onto itself, a continuous function with zero integral must vanish somewhere, and the symmetry pairs each zero c with $2 - c$. Line (6) is false. Zero integral forces at least one zero, but nothing forces the number of zeros to be odd:

$h(x) = (x-1)^2 - \frac{1}{3}$ has zero integral over $[0, 2]$ and exactly two zeros, $1 \pm \frac{1}{\sqrt{3}}$. The claim is false as well: $f(x) = x^2 - \frac{4}{3}$ has $\int_0^2 f = \frac{8}{3} - \frac{8}{3} = 0$ but $f(1) = -\frac{1}{3}$. What the argument really establishes is that h has a zero, and that zeros away from $x = 1$ come in pairs.

Question 5. Answer A.

Unpack the statement for a single function f in $\mathcal{F}$: if f is increasing, then f is bounded above unless f is a polynomial. That is: if f is increasing and f is not a polynomial, then f is bounded above. A universal implication is false exactly when some case has the hypothesis true and the conclusion false: some f that is increasing and not a polynomial, but not bounded above. That is option A. Option C is not a negation at all: it is the contrapositive of the original, and says the same thing. Option B negates the wrong part of the “unless”, and option E negates only the conclusion.

Question 6. Answer B.

A counterexample needs $f \leq g$ on $[0, 1]$ and $\int f^2 > \int g^2$. Squaring preserves $f \leq g$ only when $f \geq 0$, so any counterexample must use negative values of f . All seven pairs satisfy the hypothesis. (i) $\frac{1}{3} \leq 1$. (ii) $1 > 0$: counterexample. (iii) $\frac{1}{3} \leq \frac{1}{3}$. (iv) $\int_0^1 (2x-2)^2 dx = \frac{4}{3} > \frac{1}{3}$: counterexample. (v) $\frac{1}{5} \leq \frac{1}{3}$. (vi) $\frac{1}{3} > \frac{1}{5}$: counterexample. (vii) both integrals are $\frac{1}{3}$. Three counterexamples. Pairs (iii) and (vii) are the ones to be careful with: f is negative, but $|f|$ is no larger than $|g|$ on average, and equality still satisfies the conclusion.

Question 7. Answer E.

A counterexample needs a line that meets the curve exactly once and is not the tangent there. I: the vertical line $x = 1$ meets $y = x^2$ only at $(1, 1)$, but the tangent there has gradient 2: a counterexample. II: $y = 0$ meets $y = x^3$ only at the origin, and is the tangent there. III: $x^3 = -x$ only at $x = 0$, where the curve has gradient 0 and the line gradient -1 : a counterexample. IV: the tangent at $(1, 1)$, meeting the curve once. V: $x^3 - 12x - 16 = (x-4)(x+2)^2$, so the line is the tangent at $x = -2$ but meets the curve again at $x = 4$; the hypothesis fails. VI: $x^4 + x = x$ only at $x = 0$, where the curve's gradient is 1, the same as the line's: it is the tangent. So I and III. Meeting exactly once is how tangents to circles behave; for other curves neither property implies the other.

Question 8. Answer F.

Putting $t = 0$ in (3) gives $f(1) = -f(1)$, so (1) holds. Condition (3) says the graph has half-turn symmetry about $(1, 0)$, so the substitution $x \mapsto 2 - x$ gives $\int_1^2 f = -\int_0^1 f$, which is (2). So (3) implies both, and (1) and (2) together are necessary for (3). They are not sufficient: $f(x) = (x-1)^4 - \frac{3}{5}(x-1)^2$ has $f(1) = 0$ and $\int_0^2 f = \int_{-1}^1 (u^4 - \frac{3}{5}u^2) du = \frac{2}{5} - \frac{2}{5} = 0$, yet its graph is symmetric about $x = 1$ rather than about the point $(1, 0)$. Option A fails for $f(x) = (x-1)^2$, and option B for $f(x) = x^2 - \frac{4}{3}$.

Question 9. Answer A.

I is complete: the sign change gives a root in $(0, 1)$, and a function with positive derivative everywhere is increasing, so it cannot vanish twice. II is complete for the same reasons, phrased with graphs: an increasing function and a decreasing function agree at most once, and the comparison at 0 and 1 shows that they agree somewhere. III has a gap. The root r was found in $(0, 1)$, but nothing says that a second root s would be there too, so “ r and s are both positive” is unjustified. The step can be repaired, since $r^2 + rs + s^2 = (r + \frac{s}{2})^2 + \frac{3}{4}s^2 \geq 0$ for all real r and s , but as written the argument assumes something it has not shown.

Question 10. Answer D.

Lines (1) to (3) are correct; in (3), $x \geq 2$ gives $x^3 \geq 8$ and so $x^4 \geq 8x$. Line (4) is the first mistake: a continuous function that is positive at both ends of an interval can dip below zero in between, and its values at two points say nothing about the middle. Here the expression actually equals 0 at $x = 1$, so “positive for every x in between” is false as stated. The claim itself is true:

$x^4 - 4x + 3 = (x - 1)^2(x^2 + 2x + 3)$, and $x^2 + 2x + 3 = (x + 1)^2 + 2 > 0$. The conclusion is right, but the argument has not earned it.

Question 11. Answer D.

If $a \neq 0$, the series converges exactly when $|r| < 1$, with sum $\frac{a}{1-r}$, and this equals $2a$ exactly when $r = \frac{1}{2}$. So $|r| < 1$ is not sufficient: $a = 1$, $r = 0$ gives sum 1, not 2. It is not necessary either, because of the case $a = 0$: then every term is 0, and the series converges to $0 = 2a$ for every r , including $r = 5$. So the condition is neither necessary nor sufficient. The instinct that “ $|r| < 1$ is needed for convergence” fails only when $a = 0$, and one case is all it takes.

Question 12. Answer C.

(i) is false: for $k = 0$, $f_0(x) = x^2$ is never negative. (ii) is true with $x = 0$, where every f_k vanishes. (iii) is false for the same reason: at $x = 0$ no k makes $f_k(0) < 0$. (iv) is true: the least value of f_k is $-\frac{k^2}{4}$, which is at least -1 whenever $|k| \leq 2$. (v) is true: $f_k(x+1) - f_k(x) = 2x + 1 - k$, which is zero at $x = \frac{k-1}{2}$. (vi) is false: $f_k(x) \geq f_k(x+1)$ means $k \geq 2x + 1$, and no single x makes this hold for every k . Three statements are true. Notice that (i) and (iii) fail for different reasons, one at a single value of k and the other at a single value of x .

Question 13. Answer F.

Write $\log_a b = \frac{\ln b}{\ln a}$. Then $\log_a b > 1$ says $\frac{\ln b - \ln a}{\ln a} > 0$: the numbers $\ln b - \ln a$ and $\ln a$ have the same sign. Since $\ln$ is increasing, $\ln b - \ln a$ has the sign of $b - a$, and $\ln a$ has the sign of $a - 1$. So the condition is exactly $(b - a)(a - 1) > 0$. In words: $b > a$ when the base exceeds 1, but $b < a$ when the base lies between 0 and 1. Option A ignores the second case, since $a = \frac{1}{2}$, $b = \frac{1}{4}$ gives $\log_a b = 2$; option C is the condition for $\log_a b > 0$; option E fails for $a = \frac{1}{2}$, $b = \frac{3}{4}$, where $\log_a b \approx 0.42$.

Question 14. Answer G.

II: for a linear f the integral over $[0, 1]$ equals $f(\frac{1}{2})$, the average of the end values. III: split $[0, 1]$ into the middle half and the two outer quarters. On the middle half $f > 1$, contributing more than $\frac{1}{2}$; on the outer quarters $f > -1$, contributing more than $-\frac{1}{2}$; the total is positive. V: f is continuous and positive on $(0, 1)$ apart from one point, so its integral is positive. I fails: $f(x) = 1 - 10x(1 - x)$ has $f(0) = f(1) = 1$ but integral $1 - \frac{10}{6} < 0$. IV fails: $f(x) = 2 - 32(1 - x)^5$ is increasing, with $f(\frac{1}{2}) = 1 > 0$, but its integral is $2 - \frac{32}{6} < 0$. An increasing function can be very negative on a long stretch and only just positive at the midpoint. So II, III and V.

Question 15. Answer B.

The key fact is $\max(|x+y|, |x-y|) = |x| + |y|$: if x and y have the same sign then $|x+y| = |x| + |y|$, and otherwise $|x-y| = |x| + |y|$, and in each case the other expression is no larger. So I is equivalent to $|x| + |y| \leq 1$. V is too: for $t = |x| + |y| \geq 0$, the inequality $t^3 \leq t$ holds exactly when $t \leq 1$. IV is not: $|x+y| + |x-y| = 2\max(|x|, |y|)$, so IV says $|x| \leq 1$ and $|y| \leq 1$, which is II, and $x = y = 1$ satisfies it. III is not: $x = y = 0.6$ satisfies $x^2 + y^2 \leq 1$ but has $|x| + |y| = 1.2$. So I and V. Geometrically, I describes the square bounded by the four lines $x \pm y = \pm 1$, which is the diamond itself.

Question 16. Answer E.

For a non-constant polynomial, f' has only finitely many zeros, so near a it keeps one sign on each side. If f' changes sign at a , then f rises and then falls, or falls and then rises, so there is a local maximum or minimum; if it does not change sign, f is increasing or decreasing through a , and there is neither. So Q and S are equivalent. Both imply P , since a polynomial that changes sign at a vanishes there. R implies Q by the second derivative test. The implications stop there: $f(x) = x^3$ at $a = 0$ satisfies P but not Q , and $f(x) = x^4$ at $a = 0$ satisfies Q but not R .

Question 17. Answer C.

For $x^3 - 3x$, $x + \sin x$ and $x \sin x$, the values go below any given c at some point and above it at a later point. $x^3 - 3x$ tends to $-\infty$ on the left and $+\infty$ on the right. $x + \sin x$ lies between $x - 1$ and $x + 1$, so it does the same. $x \sin x$ is more interesting: it keeps returning to 0, but at $x = 2\pi n + \frac{3\pi}{2}$ it equals $-x$, and at $x = 2\pi n + \frac{\pi}{2}$ it equals x . So for any c we can take a of the first kind with $f(a) < c$, and then $b > a$ of the second kind with $f(b) > c$. The others fail. $1 - x$ is decreasing, so $a < b$ forces $f(a) > f(b)$. x^2 fails at $c = -1$, since nothing lies below it. $\frac{x}{1+|x|}$ stays between -1 and 1 , so it fails at $c = 2$, even though it is increasing. Three crossing functions.

Question 18. Answer B.

Translate each statement into a condition on x . P is “if $x^2 < 4$ then $x > 1$ ”, which is true when $x^2 \geq 4$ or $x > 1$: that is, $x \leq -2$ or $x > 1$. Q is “if $x \geq -2$ then $x > 1$ ”, true when $x < -2$ or $x > 1$. R is “if $x < 0$ then $x^2 > 1$ ”, true when $x \geq 0$ or $x < -1$. Now count region by region. For $x < -2$ all three hold; at $x = -2$ exactly P and R hold; for $-2 < x < -1$ only R ; for $-1 \leq x < 0$ none; for $0 \leq x \leq 1$ only R ; for $x > 1$ all three. So exactly two are true only at $x = -2$. The single point comes from the boundary: at $x = -2$ the hypothesis of P fails, since $4 < 4$ is false, so P is true; while Q needs $x < -2$ or $x > 1$ and just misses.

Question 19. Answer A.

(1): for every x in $[0, 1]$ we need $k - x^2$ to lie in $[0, 1]$. Taking $x = 1$ forces $k \geq 1$, and taking $x = 0$ forces $k \leq 1$; conversely $k = 1$ works, with $y = \sqrt{1 - x^2}$. So $\mathcal{A} = \{1\}$. (2): some single x must give $x^2 + y^2 \leq k$ for all y , the worst y being 1, so we need $x^2 + 1 \leq k$ for some x ; the best choice is $x = 0$, and $\mathcal{B}$ is the set of $k \geq 1$. (3): for each x we may choose $y = 1$, so we need $x^2 + 1 \geq k$ for every x ; the worst case is $x = 0$, and $\mathcal{C}$ is the set of $k \leq 1$. The distractors come from changing the quantifiers: “some x and some y with $x^2 + y^2 = k$ ” gives $0 \leq k \leq 2$, and “every x and every y with $x^2 + y^2 \leq k$ ” gives $k \geq 2$.

Question 20. Answer G.

I: $f(x) = 3x^2 - 2x$ works, with integral $1 - 1 = 0$. II: $f(x) = 6x(1 - x)$ works. III: $f' \leq 1$ and $f(0) = 0$ give $f(x) \leq x$ on $[0, 1]$, so $\int_0^1 f \leq \frac{1}{2}$, with equality only if $f(x) = x$ throughout; and $f(x) = x$ does satisfy all three conditions, with $f' = 1 \leq 1$. So III is possible, but only just. IV: $f(x) = x^{99}$ works, with integral $\frac{1}{100}$. V is impossible: f is increasing with $f(1) = 1$, so $f \leq 1$ on $[0, 1]$, and f is close to 0 near $x = 0$, so the integral is less than 1. So I, II, III and IV. The trap is in III: an inequality that bounds the integral can still be attained.